MILITARY SPECIFICATION

STIRRING BAR SET, MAGNETIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one set of magnetic stirring bars suitable for use with magnetic stirrer-hot plate.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-403	Plastic Molding Material, Polytetrafluoroethylene (TFE Fluorocarbon)
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-P-116	Preservation-packaging, Methods of.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

MAGNETIC MATERIAL PRODUCERS ASSOCIATION (MMPA)

Standard No. 0100 Permanent Magnet Materials

(Application for copies should be addressed to the Magnetic Material Producers Association, 1717 Howard Street, Evanston, IL 60202.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Polytetrafluoroethylene. Polytetrafluoroethylene shall be in accordance with L-P-403, type I.

3.2 Construction. Stirring bars shall be magnetic metal, coated with polytetrafluoroethylene. Bars shall be supplied with pivot rings. Rings shall be integral or removable. If integral, rings shall be centered along the longitudinal axis. Cross section of the coated stirring bar shall be round or octagonal.

3.3 Components. Set shall contain six stirring bars of the quantity and dimensions shown in Table I. Tolerance on all dimensions shall be + 1/32-inch, - 0 inch.

Table I Dimensions in inches

Quantity	Round: Diameter; or Octagonal: Distance between parallel sides	Length
2	5/16	1/2
1	5/16	1-1/2
1	3/8	1/2
1	3/8	1-1/2
1	1/2	3

3.4 Performance. Magnetic portion of stirring bars shall meet the requirements shown in Table II when tested as specified in 4.4.1.

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Table II Magnetic characteristics

Characteristic	Minimum measurement
Residual flux density	12,670 Gauss
Coercive force	640 Oersteds
Energy product	4.95 x 10^6 Gauss Oersteds

3.5 Workmanship. Workmanship shall be first class throughout. Sets shall be free from defects listed in Table V and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the magnetic metal, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table III. Unit of product shall be the set.

Table III For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table IV. Acceptance number shall be zero, and the unit of product shall be the set.

Table IV For tests

Characteristic	Requirement	Test procedure	Inspection level
Residual flux density	3.4	4.4.1	S-2
Coercive force	3.4	4.4.1	S-2
Energy product	3.4	4.4.1	S-2

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in Table V.

Table V Classification of defects

Category	Defect
Major	
101	Pivot rings cannot be secured to stirring bars (applicable to removable rings).
102	Pivot rings not permanent and centered (applicable to integral rings).
103	Coating not continuous.
104	Bars not free of deformation.
105	Component missing.
Minor	
201	Bars not free of foreign matter.

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4.3.1 Dimensional examination. The bars shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specificatior requirements.

4.4.1 Magnetism. Bars shall be tested for magnetism in accordance with Magnetic Material Producers Association Standard No. 0100.

4.5 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-ST D-129.

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packing shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One set containing six stirring bars, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Bars shall be cleaned in accordance with MIL-P-116, method C-1.

5.1.2.2 Drying. Bars shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be applied to bars.

5.1.2.4 Unit packaging. Each set of bars shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.2.5 Intermediate package. Twelve unit packages shall be packed in a box conforming to PPP-B-566, or PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each set of bars shall be clean and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged bars shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit package, intermediate and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-926-1260	STIRRING BAR SET, MAGNETIC, Laboratory Hot-Plate, Permanent Magnet, Tetrafluoroethylene Coated

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-0976

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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